AMPAK Technology Inc.

# 正基科技股份有限公司

# SPECIFICATION

**PRODUCT NAME：AP6212**

**REVISION：3.4(WEB)**

**DATE：Oct. 24th , 2025**

| Customer APPROVED | |
|---|---|
| Company | |
| Representative Signature | |

| PREPARED | REVIEW | | | | APPROVED | DCC ISSUE |
|---|---|---|---|---|---|---|
| | PM | QA | ET | | | |
| | | | PE | | | |
| | | | ME | | | |
| | | | APPROVED | | | |

正基科技股份有限公司

AMPAK Technology Inc.

# AP6212

## Data Sheet

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AMPAK Technology Inc.

# Revision History

| Date | Revision Content | Revised By | Version |
|---|---|---|---|
| 2017/04/28 | Refer to the earlier release for detailed revision history. | Richard | 1.9 |
| 2017/06/14 | - Modify FM spec.<br>- Add BT modulation 8DPSK | Richard | 2.0 |
| 2017/09/07 | - Modify Recommended Reflow Profile | Richard | 2.1 |
| 2017/09/29 | - Add Packing Dimension photo | Beth | 2.2 |
| 2018/10/08 | - Modify Operating Temperature. | Richard | 2.3 |
| 2021/01/25 | - Modify Package Information.<br>- Modify Dimension. | Richard | 2.4 |
| 2021/02/23 | - Modify BT Specification to 5.2. | Richard | 2.5 |
| 2021/05/03 | - Modify Wi-Fi received sensitivity specification | Eason | 2.6 |
| 2021/06/16 | -Modify VBAT and VDDIO supply voltage Specification | Richard | 2.7 |
| 2021/09/23 | -Modify Storage Temperature Range | Richard | 2.8 |
| 2022/11/15 | -Modify Vbat supply Range | Richard | 2.9 |
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| 2024/05/13 | - Modify Package Dimension | Ming | 3.1 |
| 2025/04/08 | - Add UART PCM Interface Description | Eason | 3.2 |
| 2025/04/08 | - Add UART Interface Description | Ali | 3.3 |
| 2025/10/24 | - Update Pin Definition | Ali | 3.4 |

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# 1. Introduction

AMPAK Technology would like to announce a low-cost and low-power consumption module which has all of the Wi-Fi, Bluetooth functionalities. The highly integrated module makes the possibilities of web browsing, VoIP, Bluetooth headsets and other applications. With seamless roaming capabilities and advanced security, also could interact with different vendors' 802.11b/g/n Access Points in the wireless LAN.

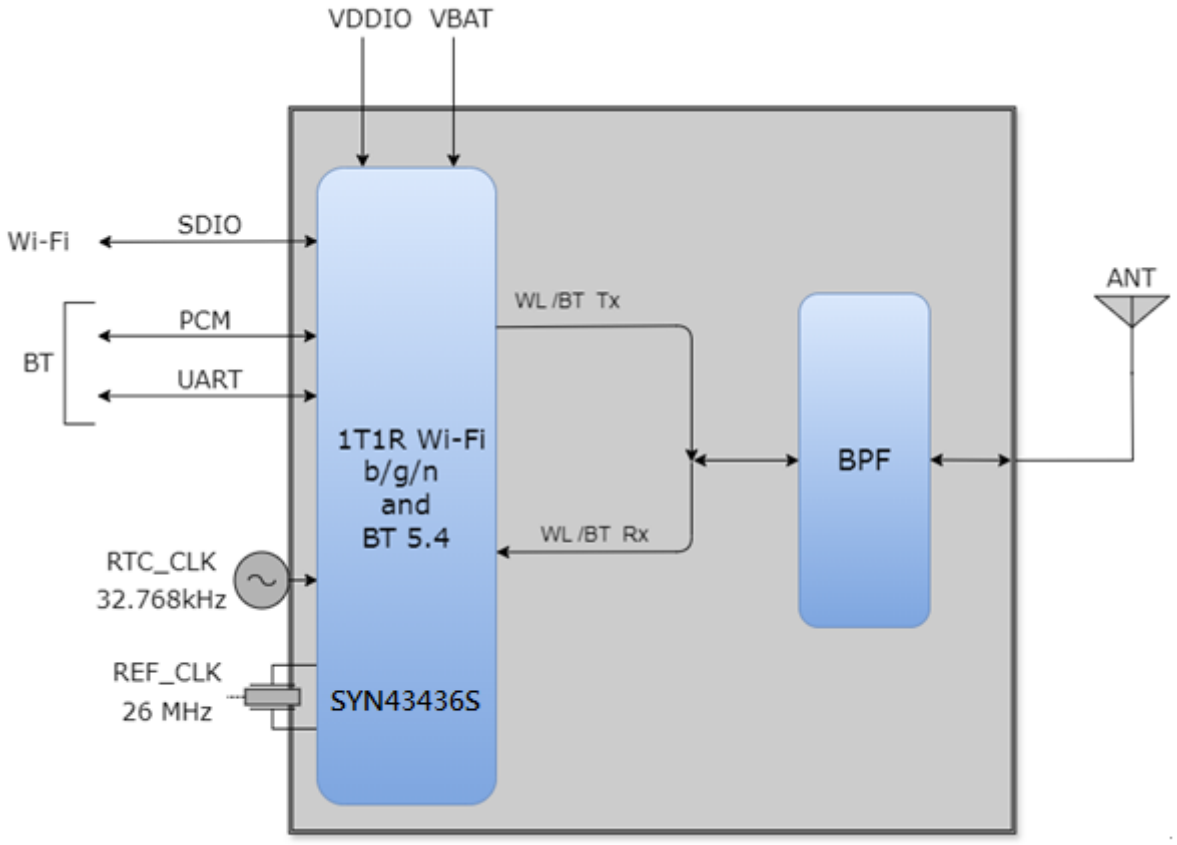
The wireless module complies with IEEE 802.11 b/g/n standard and it can achieve up to a speed of 72.2Mbps with single stream in 802.11n draft, 54Mbps as specified in IEEE 802.11g, or 11Mbps for IEEE 802.11b to connect to the wireless LAN. The integrated module provides SDIO interface for Wi-Fi, UART / I2S / PCM interface for Bluetooth.

This compact module is a total solution for a combination of Wi-Fi + BT technologies. The module is specifically developed for Smart phones and Portable devices.

# 2. Features

- 802.11b/g/n single-band radio
- Bluetooth V5.4 with integrated Class 1.5 PA and Low Energy (BLE) support
- Concurrent Bluetooth and WLAN operation
- Simultaneous BT/WLAN receive with single antenna
- WLAN host interface options:
  - SDIO v2.0 — up to 50 MHz clock rate
- BT host digital interface:
  - UART (up to 4 Mbps)
- IEEE Co-existence technologies are integrated die solution
- ECI — enhanced coexistence support, ability to coordinate BT SCO transmissions around WLAN receives

A simplified block diagram of the module is depicted in the figure below.



# 3. Deliverables

## 3.1 Deliverables

The following products and software will be part of the product.

- Module with packaging
- Evaluation Kits
- Software utility for integration, performance test.
- Product Datasheet.
- Agency certified pre-tested report with the adapter board.

## 3.2 Regulatory certifications

The product delivery is a pre-tested module, without the module level certification. For module approval, the platform's antennas are required for the certification.

# 4. General Specification

## 4.1 General Specification

| | |
|---|---|
| Model Name | AP6212 |
| Product Description | Support Wi-Fi/Bluetooth functionalities |
| Dimension | L x W : 12 x 12 (typical)mm，H:1.7 (Maximum) mm |
| Wi-Fi Interface | SDIO V2.0 |
| BT Interface | UART / PCM |
| Operating temperature [a,b] | -30°C to 85°C |
| Storage temperature | -40°C to 125°C |
| Humidity | Operating Humidity 10% to 95% Non-Condensing |

a. The operating temperature 65 to 85°C is feasible at conditional environment. Please examine the reliability on final product.

b. Functionality is guaranteed across this range of temperature. Optimal RF performance as specified in the data sheet, however, is guaranteed only for –10°C to 55°C and 3.2V < VBAT <3.8V

## 4.2 Voltages

### 4.2.1 Absolute Maximum Ratings

| Symbol | Description | Min. | Max. | Unit |
|---|---|---|---|---|
| VBAT | Input supply Voltage | -0.5 | +5.25 | V |
| VDDIO | Digital/Bluetooth/SDIO/ I/O Voltage | -0.5 | +3.8 | V |

### 4.2.2 Recommended Operating Rating

The module requires two power supplies: VBAT and VDDIO.

| | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|
| VBAT | 3.0 | 3.3 | 4.8 | V |
| VDDIO | 1.71 | 1.8 | 1.98 | V |
| | 2.97 | 3.3 | 3.63 | V |

# 5. Wi-Fi RF Specification

## 5.1 2.4GHz RF Specification

Conditions : VBAT=3.3V ; VDDIO=3.3V ; Temp:25°C

| **Feature** | **Description** |
|---|---|
| WLAN Standard | IEEE 802.11b/g/n, Wi-Fi compliant |
| Frequency Range | 2.400 GHz ~ 2.4835 GHz (2.4 GHz ISM Band) |
| Number of Channels | 2.4GHz：Ch1 ~ Ch13 |
| Modulation | 802.11b : DQPSK, DBPSK, CCK<br>802.11 g/n : OFDM /64-QAM,16-QAM, QPSK, BPSK |
| Output Power | 802.11b /11Mbps : 16 dBm ±1.5dB @ EVM ≦ -9dB |
| | 802.11g /54Mbps : 15 dBm ±1.5dB @ EVM ≦ -25dB |
| | 802.11n /65Mbps : 14 dBm ±1.5dB @ EVM ≦ -27dB |
| **Sensitivity, tolerance ± 2 dB**<br>**CCK modulation PER ≦ 8%、OFDM modulation PER ≦ 10%** | |
| 802.11n_20MHz | - MCS=0 PER @ -88 dBm, |
| | - MCS=1 PER @ -87 dBm, |
| | - MCS=2 PER @ -86 dBm, |
| | - MCS=3 PER @ -84 dBm, |
| | - MCS=4 PER @ -81 dBm, |
| | - MCS=5 PER @ -77 dBm, |
| | - MCS=6 PER @ -75 dBm, |
| | - MCS=7 PER @ -72 dBm, |
| 802.11g | - 6Mbps PER @ -90 dBm, |
| | - 9Mbps PER @ -90 dBm, |
| | - 12Mbps PER @ -88 dBm, |
| | - 18Mbps PER @ -86 dBm, |
| | - 24Mbps PER @ -83 dBm, |
| | - 36Mbps PER @ -80 dBm, |
| | - 48Mbps PER @ -77 dBm, |
| | - 54Mbps PER @ -75 dBm, |
| 802.11b | - 1Mbps PER @ -95 dBm, |
| | - 2Mbps PER @ -94 dBm, |
| | - 5.5Mbps PER @ -92 dBm, |
| | - 11Mbps PER @ -88 dBm, |

| | |
|---|---|
| Data Rate | 802.11b : 1, 2, 5.5, 11Mbps |
| | 802.11g : 6, 9, 12, 18, 24, 36, 48, 54Mbps |
| Data Rate<br>(20MHz ,Long GI,800ns) | 802.11n: 6.5, 13, 19.5, 26, 39, 52, 58.5, 65Mbps |
| Data Rate<br>(20MHz ,short GI,400ns) | 802.11n : 7.2, 14.4, 21.7, 28.9, 43.3, 57.8, 65,72.2Mbps |
| Maximum Input Level | 802.11b : -10 dBm |
| | 802.11g/n : -20 dBm |
| Antenna Reference | Small antennas with 0~2 dBi peak gain |

# 6. Bluetooth Specification

## 6.1 Bluetooth Specification

Conditions : VBAT=3.3V ; VDDIO=3.3V ; Temp:25°C

| **Feature** | **Description** | | |
|---|---|---|---|
| ***General Specification*** | | | |
| Bluetooth Standard | BDR(1Mbps)、EDR(2、3Mbps)、LE(1Mbps) | | |
| Host Interface | UART | | |
| Antenna Reference | Small antennas with 0~2 dBi peak gain | | |
| Frequency Band | 2402MHz ~ 2480MHz | | |
| Number of Channels | 79 channels | | |
| Modulation | GFSK, π/4-DQPSK, 8DPSK | | |
| ***RF Specification*** | | | |
| | **Min.** | **Typical.** | **Max.** |
| Output Power[1] | | 5 | |
| Sensitivity @ BER=0.1%<br>for GFSK (1Mbps) | | -86 dBm | |
| Sensitivity @ BER=0.01%<br>for π/4-DQPSK (2Mbps) | | -86 dBm | |
| Sensitivity @ BER=0.01%<br>for 8DPSK (3Mbps) | | -80 dBm | |
| Sensitivity @ PER=30.8%<br>for LE | | -85 dBm | |
| Maximum Input Level | GFSK (1Mbps):-20dBm | | |
| | π/4-DQPSK (2Mbps) :-20dBm | | |
| | 8DPSK (3Mbps) :-20dBm | | |

NOTE[1]：Output power can be configured by HCD firmware.

# 7. FM Specification

## 7.1 FM Specification

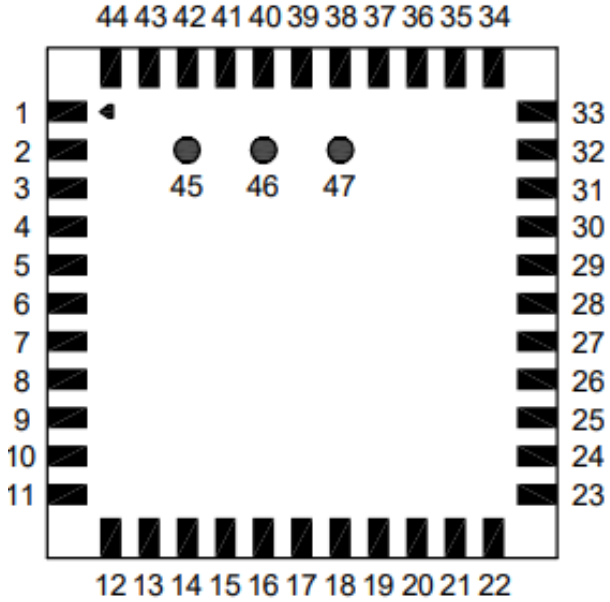
Conditions : VBAT=3.3V ; VDDIO=3.3V ; Temp:25°C

| **Feature** | **Description** | | | | | |
|---|---|---|---|---|---|---|
| ***General Specification*** | | | | | | |
| Frequency Band | 65MHz-108MHz | | | | | |
| Host Interface | HCI UART, I2S/PCM | | | | | |
| Frequency step | Minimum 10 KHz | | | | | |
| Analog Audio output load | $R_L$>30KΩ, $C_L$>20pF | | | | | |
| **Characteristics** | **Condition** | | **MIN** | **TYP** | **MAX** | **UNIT** |
| Receiver Spec. | FM only, Sensitivity, SNR ≧ 26dB | | | -5 | | dBuV |
| | Audio harmonic distortion, mono (Vin=2Mv EMF, △f=75KHz) | fmod= 1KHz | | | 0.8 | % |
| | | fmod= 3KHz | | | 0.8 | |
| | Maximum SNR (fmod=1KHz,△f=22.5KHz, BW=300Hz to 15KHz) | mono | | 69 | | dB |
| | | Stereo | | 64 | | |
| | Maximum input level, SNR > 26dB | | | | 107 | dBuV |

# 8. Pin Assignments

## 8.1 Pin Outline

<TOP VIEW>



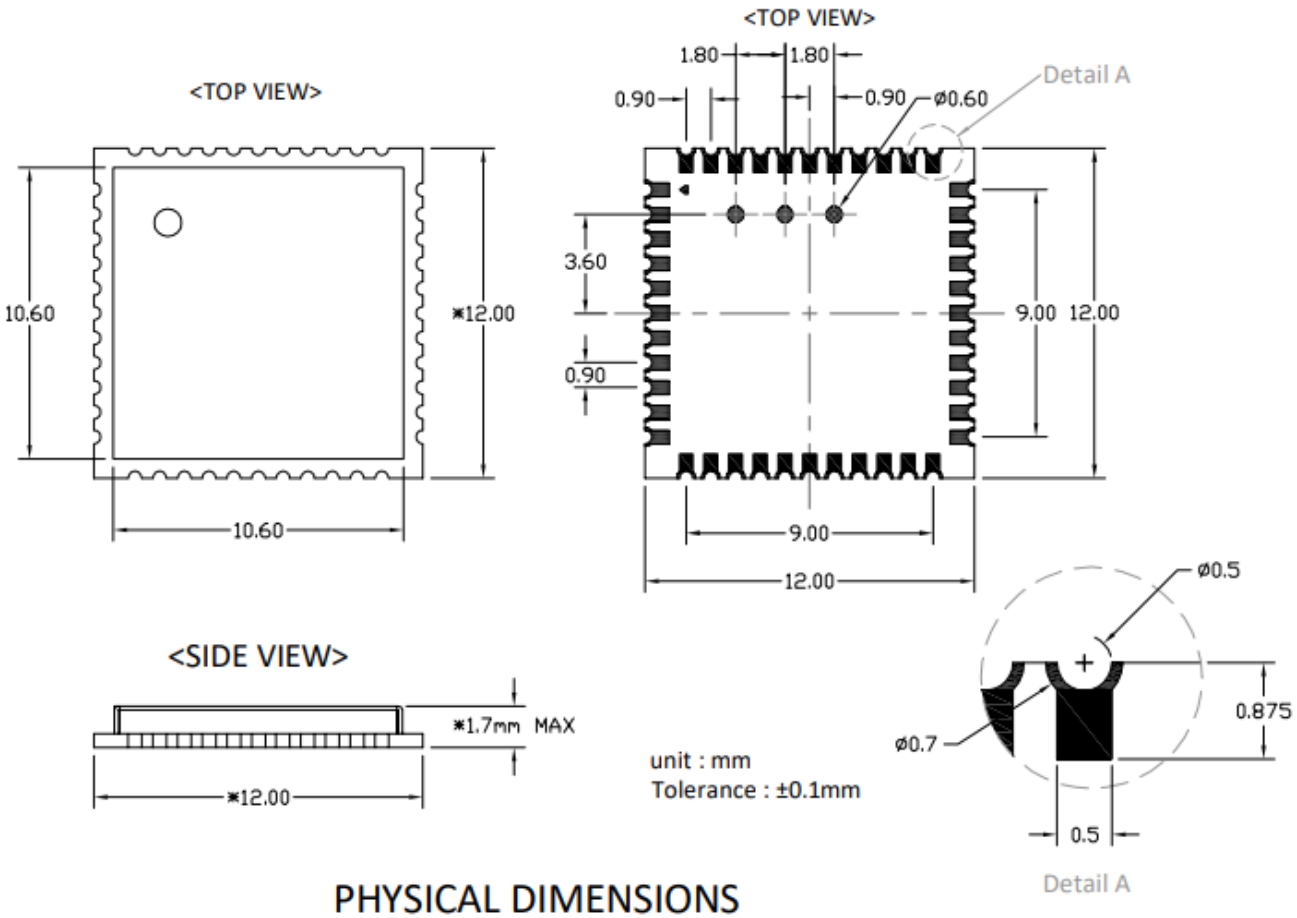

## 8.2 Pin Definition

| NO | Name | Type | Description |
|---|---|---|---|
| **1** | GND | — | Ground connections |
| **2** | WL_BT_ANT | I/O | RF I/O port |
| **3** | GND | — | Ground connections |
| **4** | FM_RX | I | FM radio RF input antenna port |
| **5** | NC | — | Floating (Don’t connected to ground) |
| **6** | BT_WAKE | I | HOST wake-up Bluetooth device |
| **7** | BT_HOST_WAKE | O | Bluetooth device to wake-up HOST |
| **8** | NC | — | Floating (Don’t connected to ground) |
| **9** | VBAT | P | Main power voltage source input |
| **10** | XTAL_IN | I | Crystal input |
| **11** | XTAL_OUT | O | Crystal output |
| **12** | WL_REG_ON | I | Internal regulators power enable/disable |
| **13** | WL_HOST_WAKE | O | WLAN to wake-up HOST |
| **14** | SDIO_DATA_2 | I/O | SDIO data line 2 |

| | | | |
|---|---|---|---|
| **15** | SDIO_DATA_3 | I/O | SDIO data line 3 |
| **16** | SDIO_DATA_CMD | I/O | SDIO command line |
| **17** | SDIO_DATA_CLK | I/O | SDIO clock line |
| **18** | SDIO_DATA_0 | I/O | SDIO data line 0 |
| **19** | SDIO_DATA_1 | I/O | SDIO data line 1 |
| **20** | GND | — | Ground connections |
| **21** | VIN_LDO_OUT | P | Internal Buck voltage generation pin |
| **22** | VDDIO | P | I/O Voltage supply input |
| **23** | VIN_LDO | P | Internal Buck voltage generation pin |
| **24** | LPO | I | External Low Power Clock input (32.768KHz) |
| **25** | PCM_OUT | O | PCM Data output |
| **26** | PCM_CLK | I/O | PCM clock |
| **27** | PCM_IN | I | PCM data input |
| **28** | PCM_SYNC | I/O | PCM sync signal |
| **29** | NC | — | Floating (Don’t connected to ground) |
| **30** | NC | — | Floating (Don’t connected to ground) |
| **31** | GND | — | Ground connections |
| **32** | NC | — | Floating (Don’t connected to ground) |
| **33** | GND | — | Ground connections |
| **34** | BT_RST_N | I | Low asserting reset for Bluetooth core |
| **35** | NC | — | Floating (Don’t connected to ground) |
| **36** | GND | — | Ground connections |
| **37** | NC | — | Floating (Don’t connected to ground) |
| **38** | NC | — | Floating (Don’t connected to ground) |
| **39** | GPIO2 | I/O | WIFI_GPIO2 |
| **40** | GPIO1 | I/O | WIFI_GPIO1 |
| **41** | UART_RTS_N | O | Bluetooth/FM UART interface |
| **42** | UART_TXD | O | Bluetooth/FM UART interface |
| **43** | UART_RXD | I | Bluetooth/FM UART interface |
| **44** | UART_CTS_N | I | Bluetooth/FM UART interface |
| **45** | TP1 | O | FM Analog AUDIO left output |
| **46** | TP2 | O | FM Analog AUDIO right output |
| **47** | TP3 (NC) | — | Floating (Don’t connected to ground) |

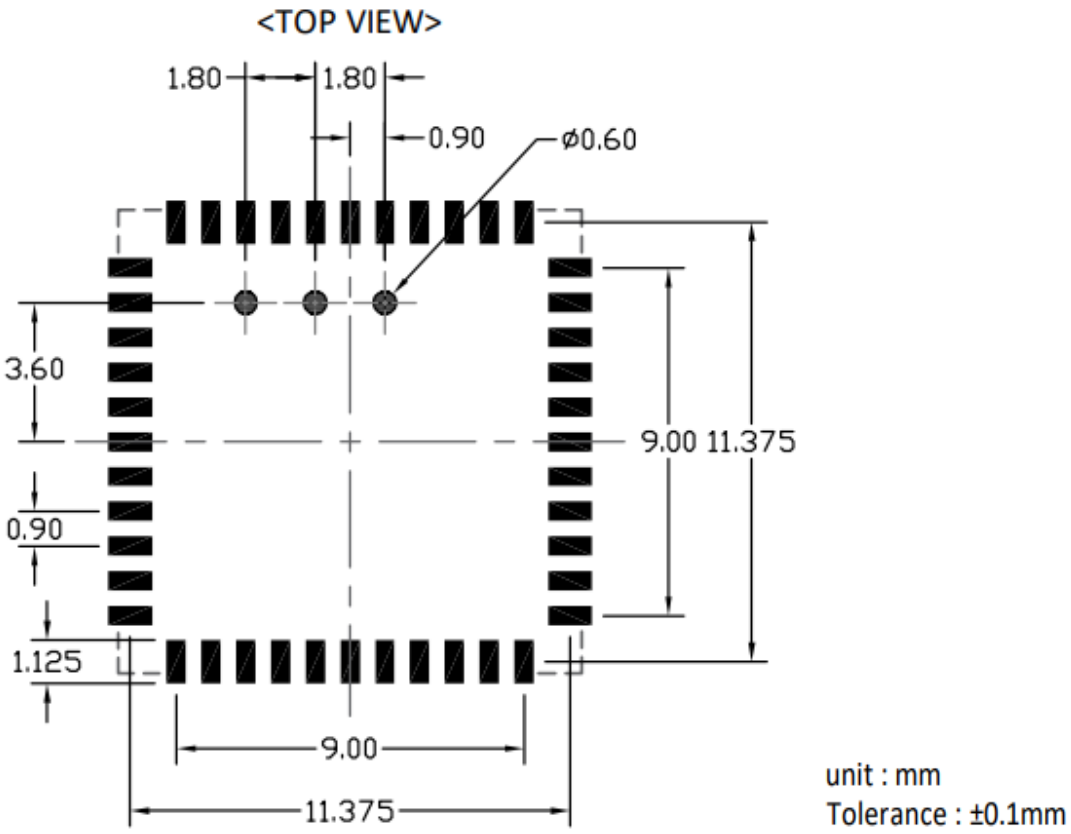
# 9. Dimensions

## 9.1 Physical Dimensions

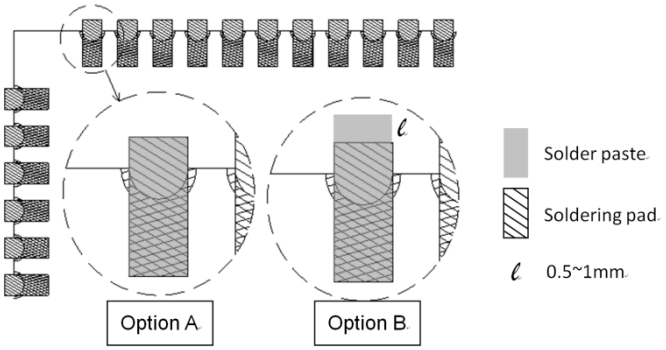



PHYSICAL DIMENSIONS

## 9.2 Layout Recommendation




- Solder paste layer design is generally the same as recommended footprint.
  (錫膏層設計通常建議和焊墊尺寸相同)
- If soldering quality with good wetting on upright side is essential for PQC, how to optimize the aperture design in the stencil to adjust the amount of solder paste would be crucial.
  In addition, a kind of stencil design with stepped thickness in partial area would be considered if the thickness of stencil is about 0.1mm or thinner. Please optimize the stencil design by manufacture engineer or contact AMPAK FAE for assistance.
  (如果模組吃錫品質考量側面爬錫，如何優化鋼網開孔設計以調整適當的錫膏量是非常重要的。尤其鋼網的厚度大約是 0.1mm 或更薄時，可考慮局部加厚鋼網的設計。請諮詢製程工程師以優化鋼網的設計,或是聯絡正基科技技術支持團隊).



# 10. External clock reference

External LPO signal characteristics

| Parameter | Specification | Units |
|---|---|---|
| Nominal input frequency | 32.768 | kHz |
| Frequency accuracy | ±30 | ppm |
| Duty cycle | 30 - 70 | % |
| Module input Signal Level | 400~3300 | mV, p-p |
| Signal type | Square-wave | - |
| Input impedance | >100k<br><5 | Ω<br>pF |
| Clock jitter (integrated over 300Hz – 15KHz) | <1 | Hz |

External Ref_CLK signal characteristics

| No. | Item | Symb. | Electrical Specification | | | | Remark |
|---|---|---|---|---|---|---|---|
| | | | Min. | Type | Max. | Units | |
| 1 | Nominal Frequency | F0 | | 26.00000 | | MHz | |
| 2 | Mode of Vibration | | | Fundamental | | | |
| 3 | Frequency Tolerance | ΔF/F0 | -10 | - | 10 | ppm | at 25℃±3℃ |
| 4 | Operating Temperature Range | $T_{OPR}$ | -30 | - | 85 | ℃ | |
| 5 | Frequency Stability | TC | -10 | - | 10 | ppm | |
| 6 | Storage Temperature | $T_{STG}$ | -55 | - | 125 | ℃ | |
| 7 | Load capacitance | CL | - | 16 | | pF | |
| 8 | Equivalent Series Resistance | ESR | - | - | 50 | Ω | |
| 9 | Drive Level | DL | - | 100 | 200 | μW | |
| 10 | Insulation Resistance | IR | 500 | - | - | MΩ | At 100$V_{DC}$ |
| 11 | Shunt Capacitance | C0 | - | - | 3 | pF | |
| 12 | Aging Per Year | Fa | -2 | - | 2 | ppm | First Year |

## 10.1 SDIO Pin Description

The module supports SDIO version 2.0 for 4-bit modes (100 Mbps), and high speed 4-bit (50 MHz clocks – 200 Mbps). It has the ability to stop the SDIO clock and map the interrupt signal into a GPIO pin. This 'out-of-band' interrupt signal notifies the host when the WLAN device wants to turn on the SDIO interface. The ability to force the control of the gated clocks from within the WLAN chip is also provided.

- Function 0 Standard SDIO function (Max BlockSize / ByteCount = 32B)
- Function 1 Backplane Function to access the internal System On Chip (SOC) address space (Max BlockSize / ByteCount = 64B)
- Function 2 WLAN Function for efficient WLAN packet transfer through DMA (Max BlockSize/ByteCount=512B)

SDIO Pin Description

| **SD 4-Bit Mode** | |
|---|---|
| DATA0 | Data Line 0 |
| DATA1 | Data Line 1 or Interrupt |
| DATA2 | Data Line 2 or Read Wait |
| DATA3 | Data Line 3 |
| CLK | Clock |
| CMD | Command Line |

# 11. Host Interface Timing Diagram

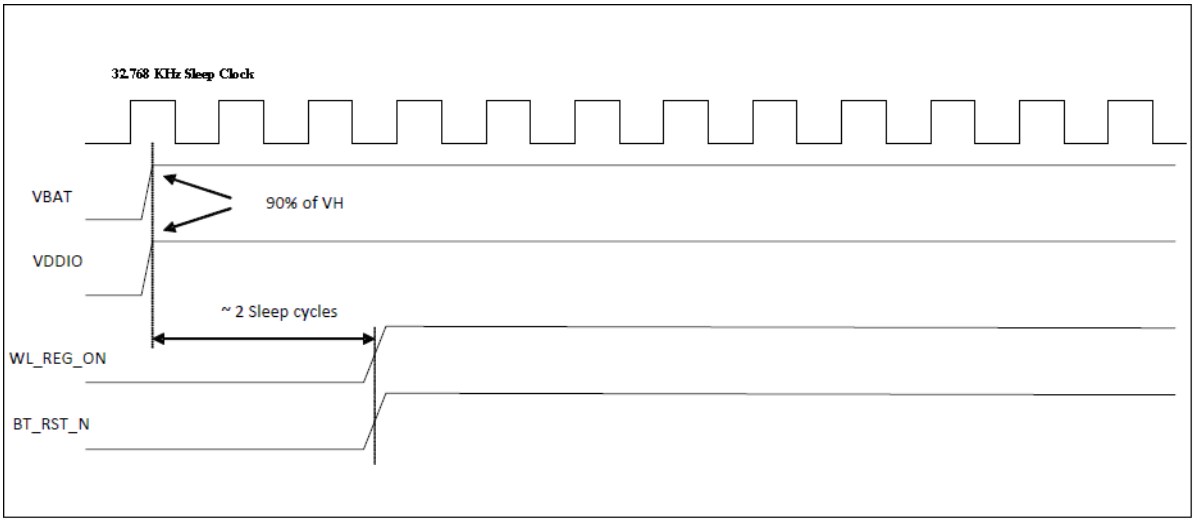
## 11.1 Power-up Sequence Timing Diagram

The module has signals that allow the host to control power consumption by enabling or disabling the Bluetooth, WLAN and internal regulator blocks. These signals are described below.

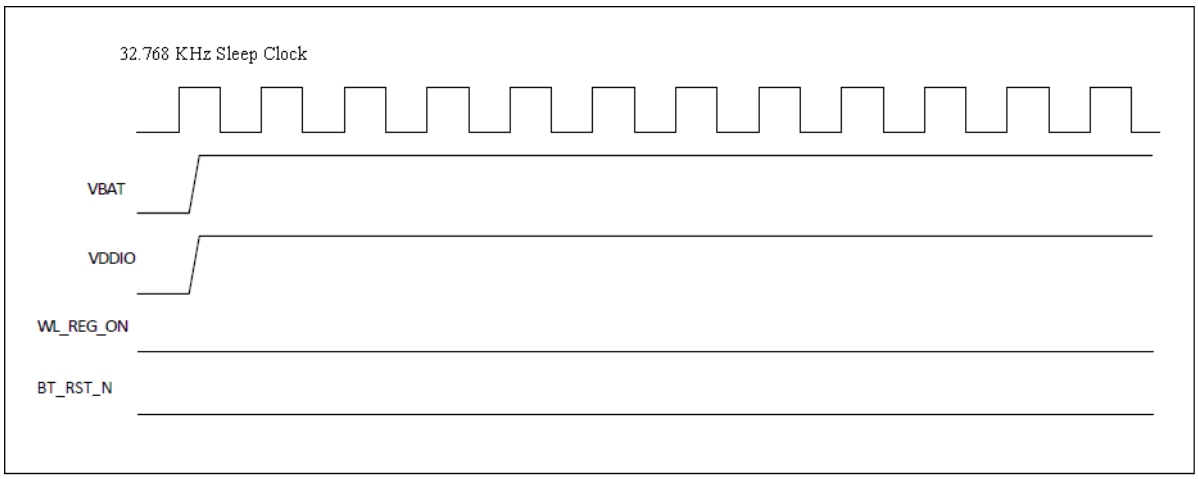
Additionally, diagrams are provided to indicate proper sequencing of the signals for carious operating states. The timing value indicated are minimum required values: longer delays are also acceptable.

※ WL_REG_ON: Used by the PMU to power up the WLAN section. When this pin is high, the regulators are enabled and the WLAN section is out of reset. When this pin is low the WLAN section is in reset.

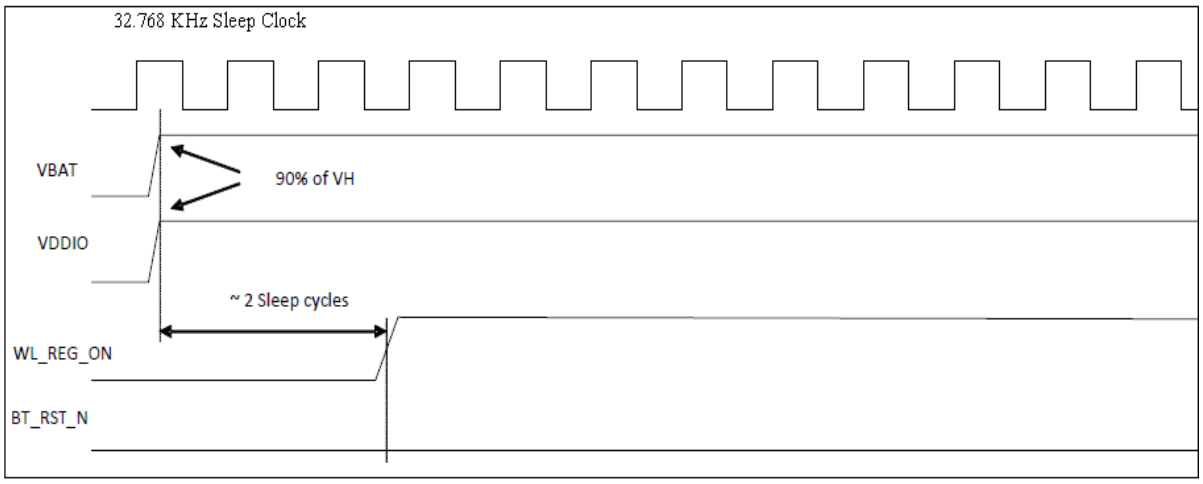
※ BT_RST_N: Low asserting reset for Bluetooth and FM only. This pin has no effect on WLAN and does not control any PMU functions. This pin must be driven high or low (not left floating).



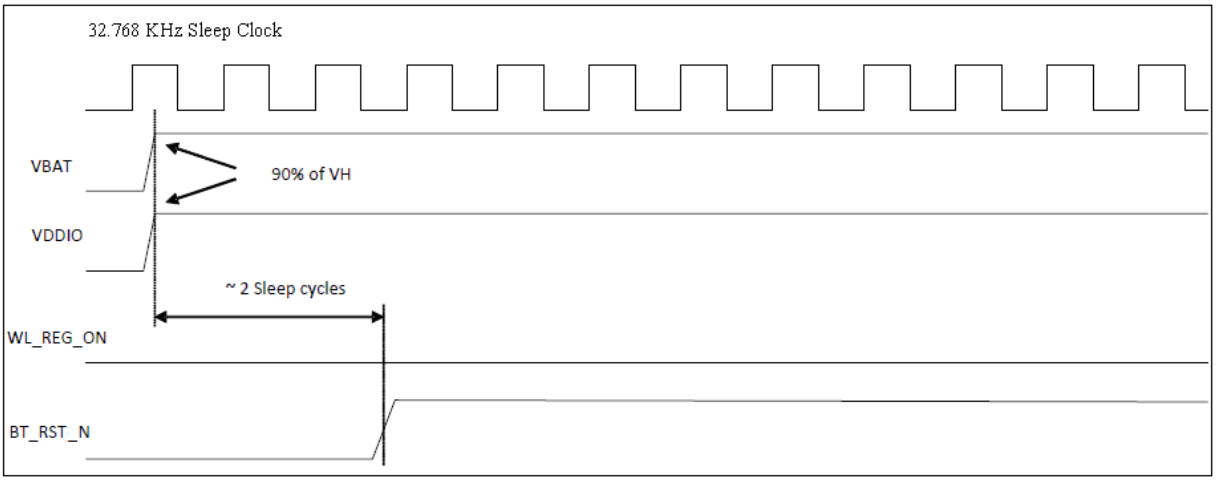

WLAN=ON, Bluetooth=ON




WLAN=OFF, Bluetooth=OFF

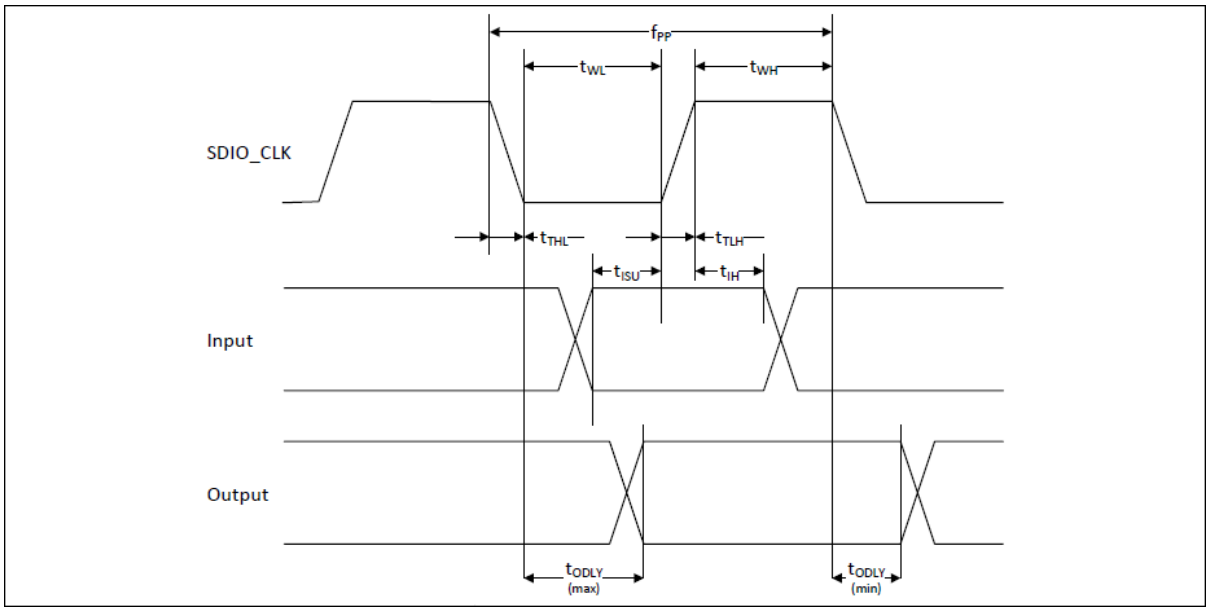



WLAN=ON, Bluetooth=OFF




WLAN=OFF, Bluetooth=ON

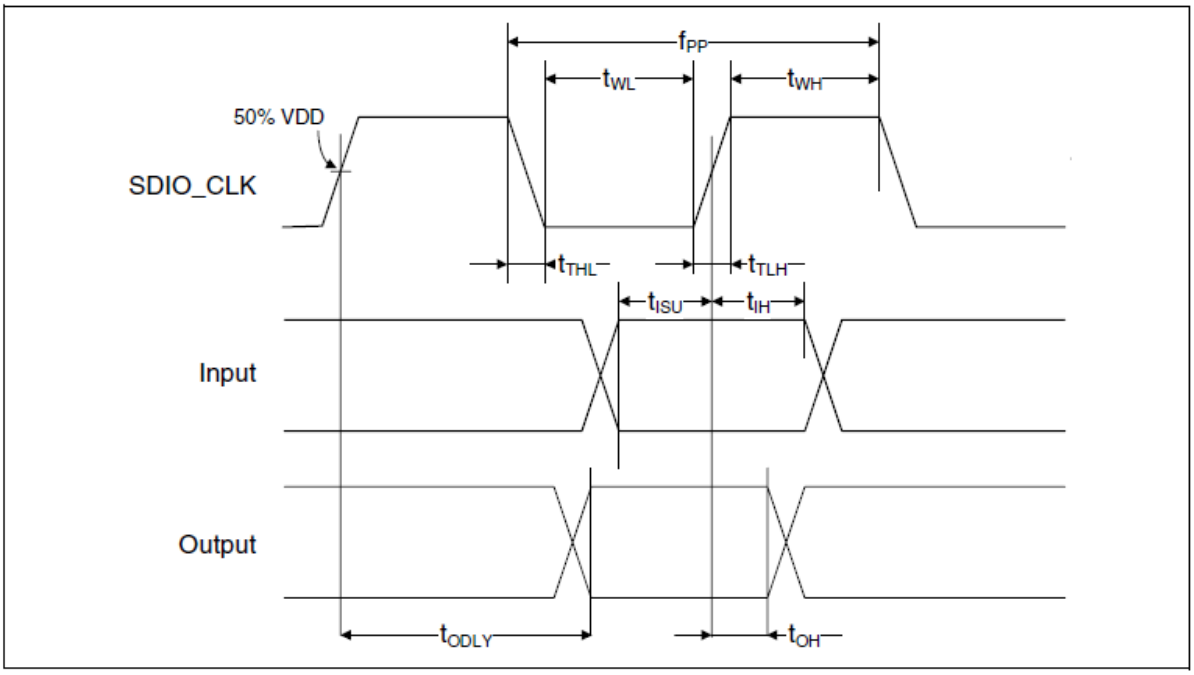
# 11.2 SDIO Default Mode Timing Diagram




| Parameter | Symbol | Minimum | Typical | Maximum | Unit |
|---|---|---|---|---|---|
| ***SDIO CLK (All values are refferred to minimum VIH and maximum VIL[b] )*** | | | | | |
| Frequency-Data Transfer mode | fPP | 0 | - | 25 | MHz |
| Frequency-Identification mode | fOD | 0 | - | 400 | kHz |
| Clock low time | tWL | 10 | - | - | ns |
| Clock high time | tWH | 10 | - | - | ns |
| Clock rise time | tTLH | - | - | 10 | ns |
| Clock low time | tTHL | - | - | 10 | ns |
| ***Inputs: CMD, DAT (referenced to CLK)*** | | | | | |
| Input setup time | tISU | 5 | - | - | ns |
| Input hold time | tIH | 5 | - | - | ns |
| ***Outputs: CMD, DAT (referenced to CLK)*** | | | | | |
| Output delay time – Data Transfer mode | tODLY | 0 | - | 14 | ns |
| Output delay time – Identification mode | tODLY | 0 | - | 50 | ns |

a. Timing is based on CL ≤ 40pF load on CMD and Data.
b. min(Vih) = 0.7 x VDDIO and max(Vil) = 0.2 x VDDIO.

# 11.3 SDIO High Speed Mode Timing Diagram




| Parameter | Symbol | Minimum | Typical | Maximum | Unit |
|---|---|---|---|---|---|
| ***SDIO CLK (All values are refferred to minimum VIH and maximum VIL[b] )*** | | | | | |
| Frequency-Data Transfer mode | fPP | 0 | - | 50 | MHz |
| Frequency-Identification mode | fOD | 0 | - | 400 | kHz |
| Clock low time | tWL | 7 | - | - | ns |
| Clock high time | tWH | 7 | - | - | ns |
| Clock rise time | tTLH | - | - | 3 | ns |
| Clock low time | tTHL | - | - | 3 | ns |
| ***Inputs: CMD, DAT (referenced to CLK)*** | | | | | |
| Input setup time | tISU | 6 | - | - | ns |
| Input hold time | tIH | 2 | - | - | ns |
| ***Outputs: CMD, DAT (referenced to CLK)*** | | | | | |
| Output delay time – Data Transfer mode | tODLY | - | - | 14 | ns |
| Output hold time | tOH | 2.5 | - | - | ns |
| Total system capacitance (each line) | CL | - | - | 40 | pF |

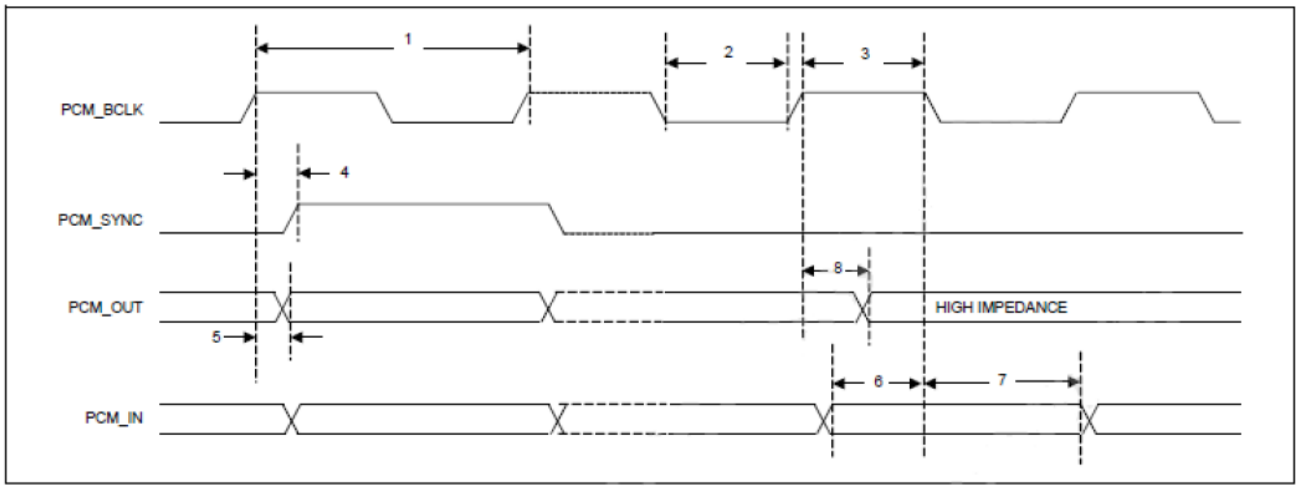
a. Timing is based on CL ≤ 40pF load on CMD and Data.
b. min(Vih) = 0.7 x VDDIO and max(Vil) = 0.2 x VDDIO.

# 11.4 PCM Interface Description

The PCM Interface on the AP6212 can connect to linear PCM Codec devices in master or slave mode. In master mode, the AP6212 generates the PCM_CLK and PCM_SYNC signals, and in slave mode, these signals are provided by another master on the PCM interface and are inputs to the AP6212.The configuration of the PCM interface may be adjusted by the host through the use of vendor-specific HCI commands.

## Short Frame Sync, Master Mode

**PCM Timing Diagram (Short Frame Sync, Master Mode)**



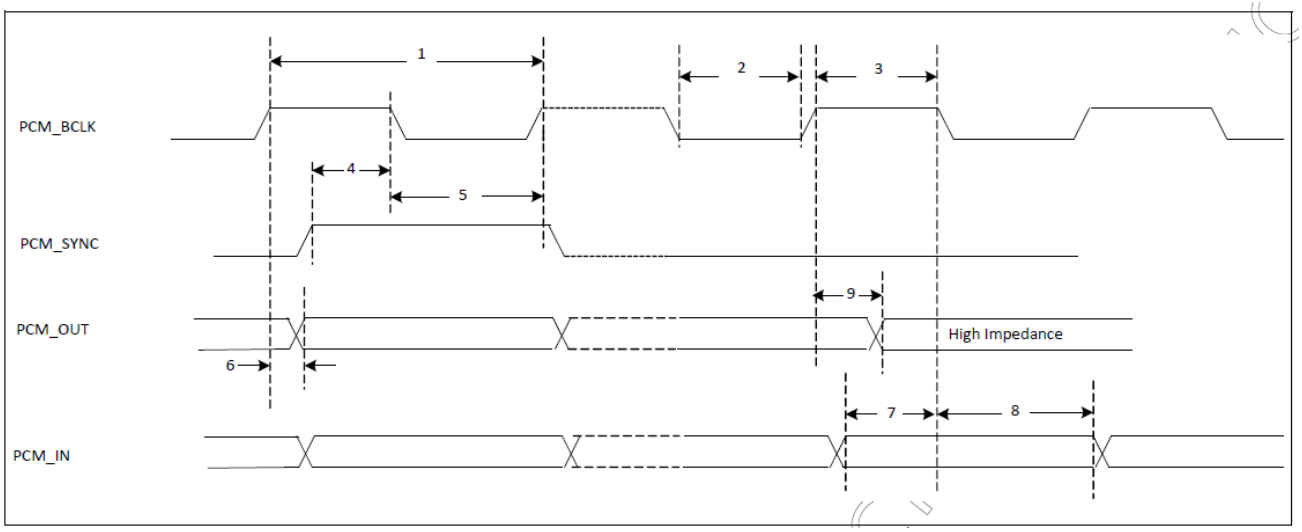

***PCM Interface Timing Specifications (Short Frame Sync, Master Mode)***

| *Ref No.* | *Characteristics* | *Minimum* | *Typical* | *Maximum* | *Unit* |
|---|---|---|---|---|---|
| 1 | PCM bit clock frequency | – | – | 12 | MHz |
| 2 | PCM bit clock low | 41 | – | – | ns |
| 3 | PCM bit clock high | 41 | – | – | ns |
| 4 | PCM_SYNC delay | 0 | – | 25 | ns |
| 5 | PCM_OUT delay | 0 | – | 25 | ns |
| 6 | PCM_IN setup | 8 | – | – | ns |
| 7 | PCM_IN hold | 8 | – | – | ns |
| 8 | Delay from rising edge of PCM_BCLK during last bit period to PCM_OUT becoming high impedance | 0 | – | 25 | ns |

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# Short Frame Sync, Slave Mode

### PCM Timing Diagram (Short Frame Sync, Slave Mode)

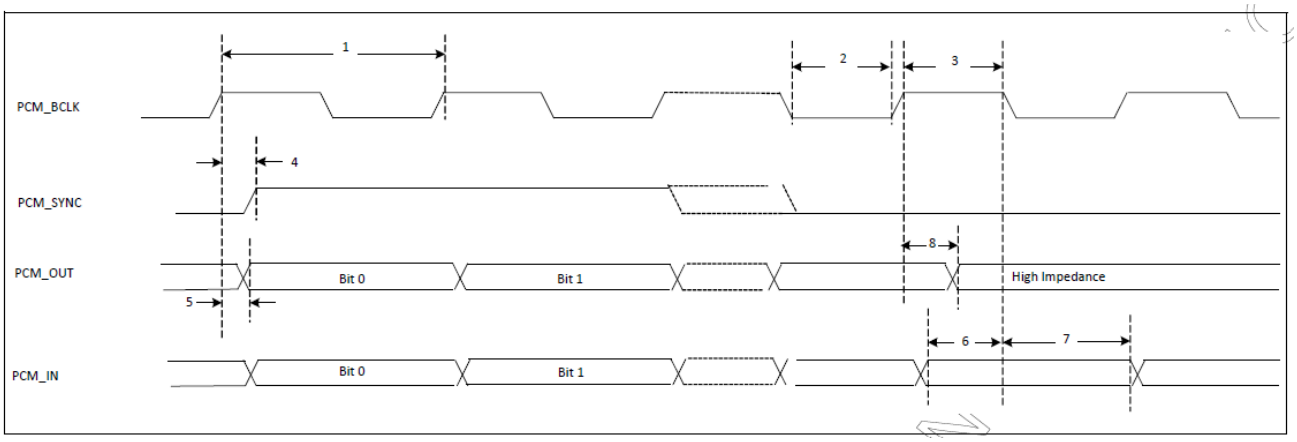



### *PCM Interface Timing Specifications (Short Frame Sync, Slave Mode)*

| *Ref No.* | *Characteristics* | *Minimum* | *Typical* | *Maximum* | *Unit* |
|---|---|---|---|---|---|
| 1 | PCM bit clock frequency | – | – | 12 | MHz |
| 2 | PCM bit clock low | 41 | – | – | ns |
| 3 | PCM bit clock high | 41 | – | – | ns |
| 4 | PCM_SYNC setup | 8 | – | – | ns |
| 5 | PCM_SYNC hold | 8 | – | – | ns |
| 6 | PCM_OUT delay | 0 | – | 25 | ns |
| 7 | PCM_IN setup | 8 | – | – | ns |
| 8 | PCM_IN hold | 8 | – | – | ns |
| 9 | Delay from rising edge of PCM_BCLK during last bit period to PCM_OUT becoming high impedance | 0 | – | 25 | ns |

# Long Frame Sync, Master Mode

### PCM Timing Diagram (Long Frame Sync, Master Mode)

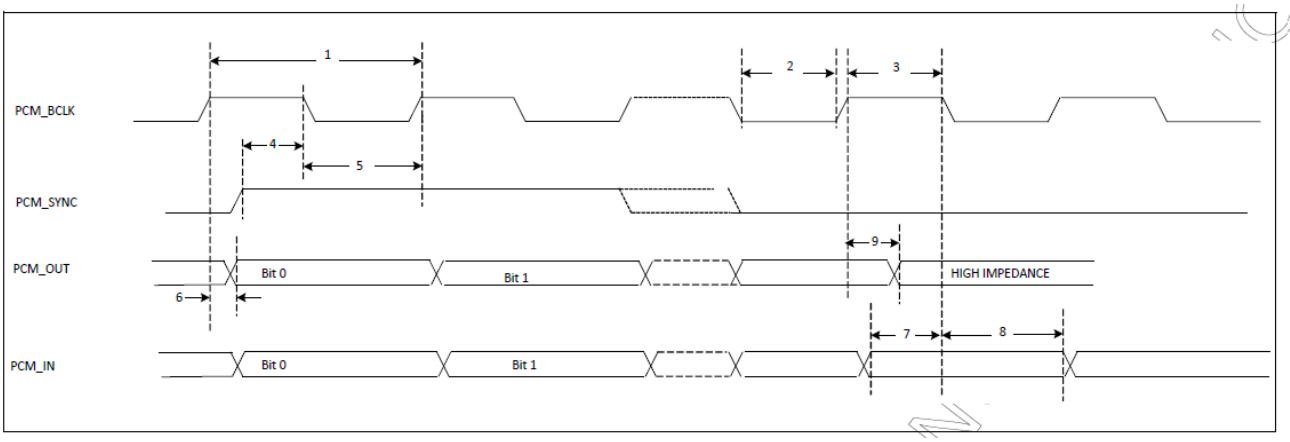



### *PCM Interface Timing Specifications (Long Frame Sync, Master Mode)*

| *Ref No.* | *Characteristics* | *Minimum* | *Typical* | *Maximum* | *Unit* |
|---|---|---|---|---|---|
| 1 | PCM bit clock frequency | – | – | 12 | MHz |
| 2 | PCM bit clock low | 41 | – | – | ns |
| 3 | PCM bit clock high | 41 | – | – | ns |
| 4 | PCM_SYNC delay | 0 | – | 25 | ns |
| 5 | PCM_OUT delay | 0 | – | 25 | ns |
| 6 | PCM_IN setup | 8 | – | – | ns |
| 7 | PCM_IN hold | 8 | – | – | ns |
| 8 | Delay from rising edge of PCM_BCLK during last bit period to PCM_OUT becoming high impedance | 0 | – | 25 | ns |

# Long Frame Sync, Slave Mode

### PCM Timing Diagram (Long Frame Sync, Slave Mode)




### *PCM Interface Timing Specifications (Long Frame Sync, Slave Mode)*

| *Ref No.* | *Characteristics* | *Minimum* | *Typical* | *Maximum* | *Unit* |
|---|---|---|---|---|---|
| 1 | PCM bit clock frequency | – | – | 12 | MHz |
| 2 | PCM bit clock low | 41 | – | – | ns |
| 3 | PCM bit clock high | 41 | – | – | ns |
| 4 | PCM_SYNC setup | 8 | – | – | ns |
| 5 | PCM_SYNC hold | 8 | – | – | ns |
| 6 | PCM_OUT delay | 0 | – | 25 | ns |
| 7 | PCM_IN setup | 8 | – | – | ns |
| 8 | PCM_IN hold | 8 | – | – | ns |
| 9 | Delay from rising edge of PCM_BCLK during last bit period to PCM_OUT becoming high impedance | 0 | – | 25 | ns |

## 11.5 UART Interface Description

The AP6212 shares a single UART for Bluetooth and FM. The UART is a standard 4-wire interface (RX, TX, RTS, and CTS) with adjustable baud rates from 9600 bps to 4.0 Mbps. The interface features an automatic baud rate detection capability that returns a baud rate selection. Alternatively, the baud rate may be selected through a vendor-specific UART HCI command.

UART has a 1040-byte receive FIFO and a 1040-byte transmit FIFO to support EDR. Access to the FIFOs is conducted through the AHB interface through either DMA or the CPU. The UART supports the Bluetooth UART HCI specification: H4, a custom Extended H4, and H5. The default baud rate is 115.2 Kbaud.

The UART supports the 3-wire H5 UART transport, as described in the Bluetooth specification (Three-wire UART Transport Layer). Compared to H4, the H5 UART transport reduces the number of signal lines required by eliminating the CTS and RTS signals.

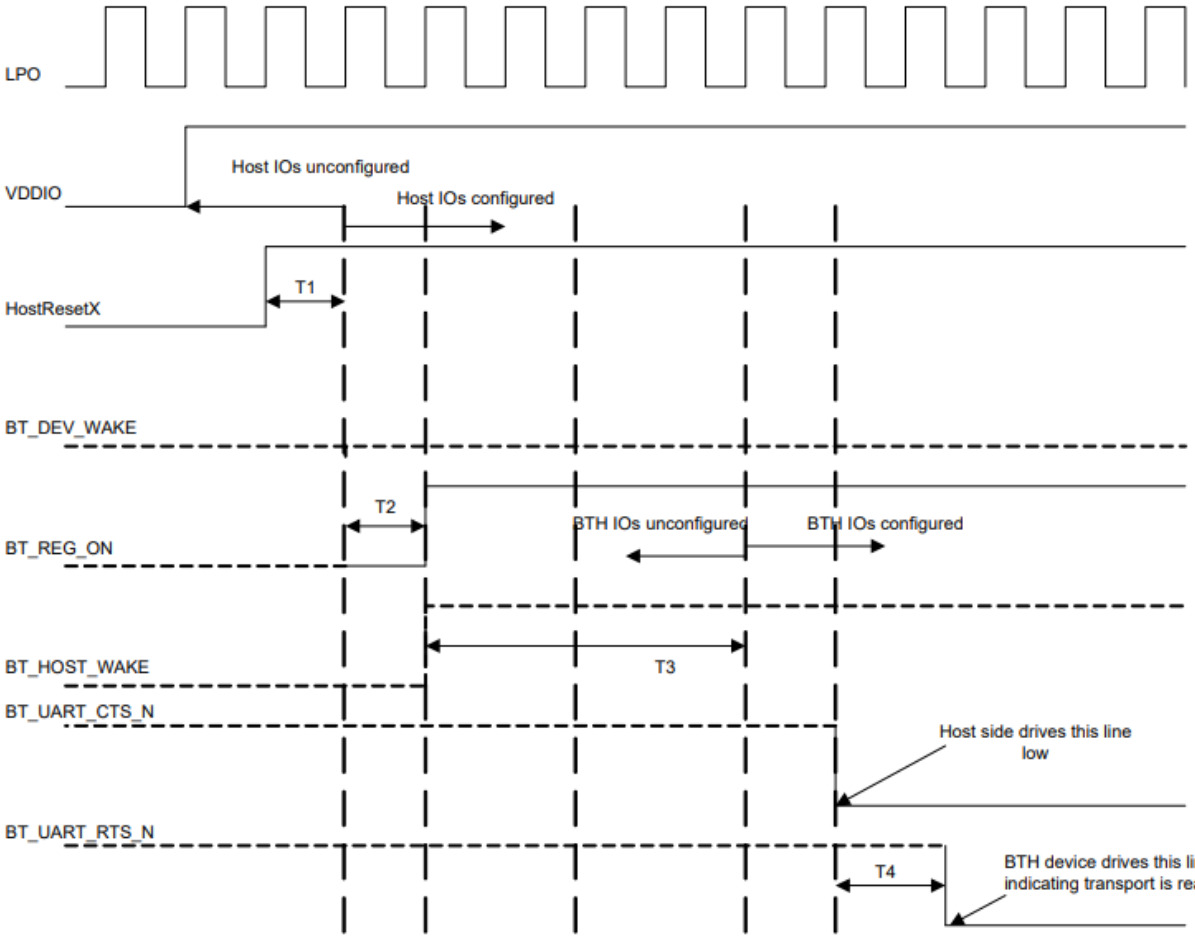
The AP6212 UART can perform XON/XOFF flow control and includes hardware support for the Serial Line Input Protocol (SLIP). It can also perform wake-on activity. For example, activity on the RX or CTS inputs can wake the chip from a sleep state.

Normally, the UART baud rate is set by a configuration record downloaded after device reset, or by automatic baud rate detection, and the host does not need to adjust the baud rate. Support for changing the baud rate during normal HCI UART operation is included through a vendor-specific command that allows the host to adjust the contents of the baud rate registers. The AP6212 UARTs operate correctly with the host UART as long as the combined baud rate error of the two devices is within ±2%.

***Example of Common Baud Rates***

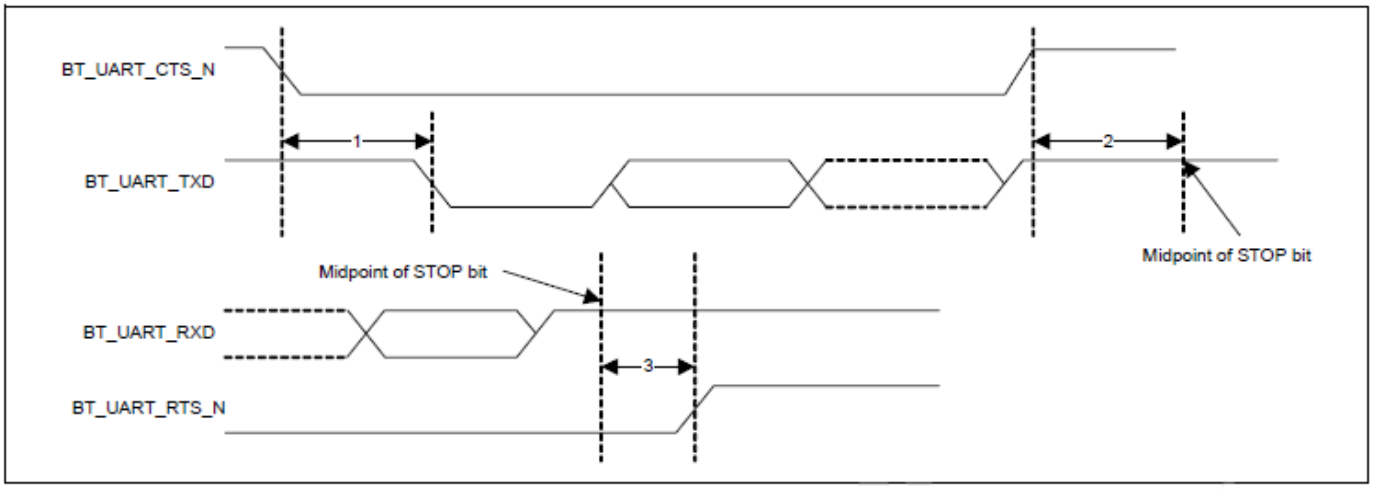
| Desired Rate | Actual Rate | Error (%) |
|---|---|---|
| 4000000 | 4000000 | 0.00 |
| 3692000 | 3692308 | 0.01 |
| 3000000 | 3000000 | 0.00 |
| 2000000 | 2000000 | 0.00 |
| 1500000 | 1500000 | 0.00 |
| 1444444 | 1454544 | 0.70 |
| 921600 | 923077 | 0.16 |
| 460800 | 461538 | 0.16 |
| 230400 | 230796 | 0.17 |
| 115200 | 115385 | 0.16 |
| 57600 | 57692 | 0.16 |
| 38400 | 38400 | 0.00 |
| 28800 | 28846 | 0.16 |
| 19200 | 19200 | 0.00 |
| 14400 | 14423 | 0.16 |
| 9600 | 9600 | 0.00 |

## Startup signaling sequence




Notes :
T1 is the time for Host to settle it's IOs after a reset.
T2 is the time for Host to drive BT_REG_ON high after the Host IOs are configured.
T3 is the time for BTH (Bluetooth) device to settle its IOs after a reset and reference clock settling time has elapsed.
T4 is the time for BTH device to drive BT_UART_RTS_N low after the Host drives BT_UART_CTS_N low. This assumes the BTH device has already completed initialization.

## UART Timing

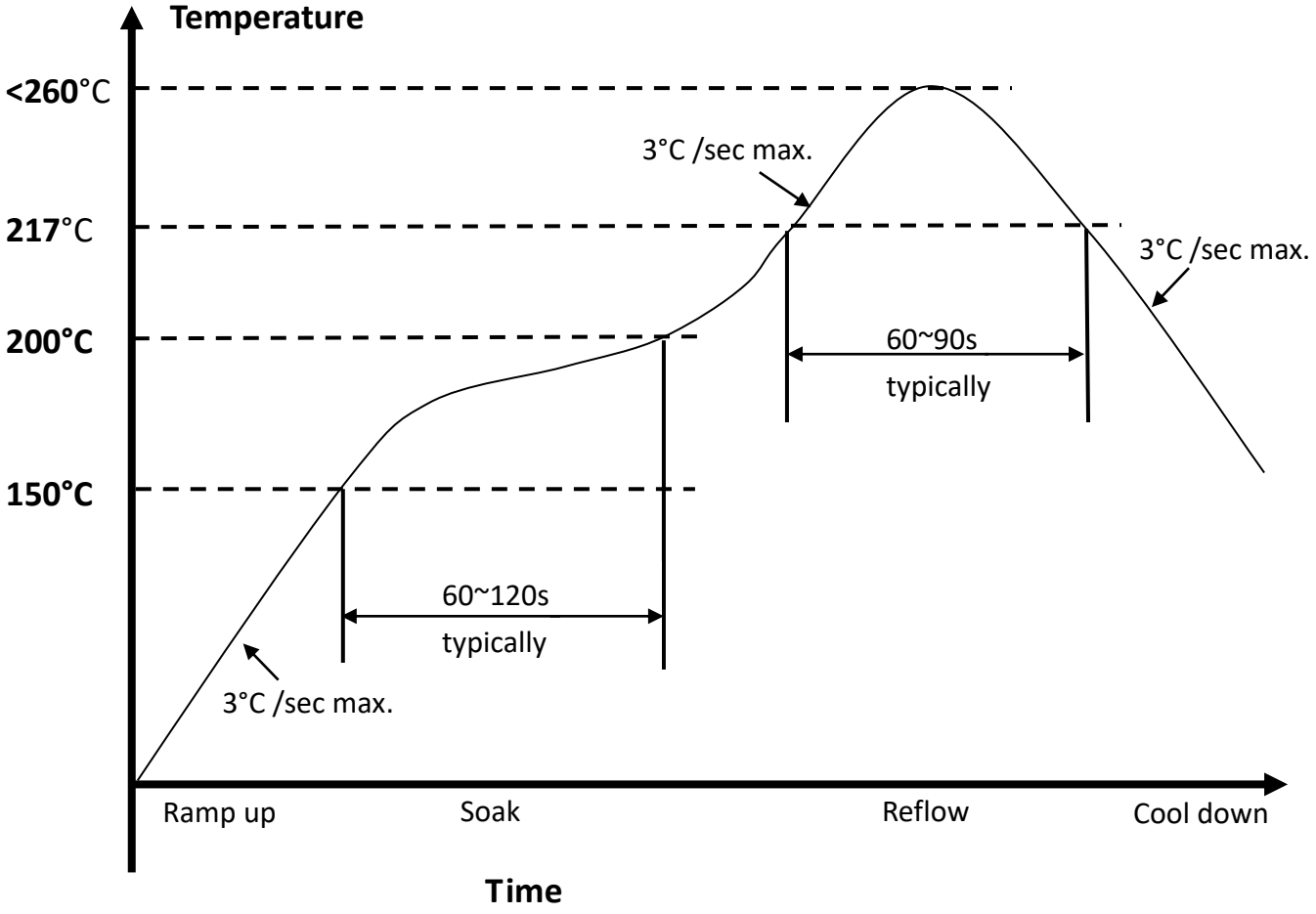



## UART Timing Specifications

| Ref | Characteristics | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| 1 | Delay time, BT_UART_CTS_N low to BT_UART_TXD valid | – | – | 1.5 | Bit periods |
| 2 | Setup time, BT_UART_CTS_N high before midpoint of stop bit | – | – | 0.5 | Bit periods |
| 3 | Delay time, midpoint of stop bit to BT_UART_RTS_N high | – | – | 0.5 | Bit periods |

# 12. Recommended Reflow Profile

Referred to IPC/JEDEC standard.
Peak Temperature : <250°C
Number of Times : ≤2 times




1. Referred to IPC/JEDEC standard
2. Peak Temperature : <260°C
3. Cycle of Reflow : 2 times max.
4. Adding Nitrogen (N2) to implement 2000ppm or less of oxygen concentration during reflow process is recommended.
5. If the shelf time is exceeded, be sure baking step to remove the moisture from the component

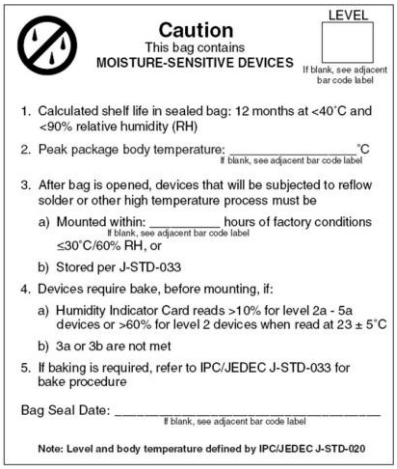
# 13. Package Information

## 13.1 Label

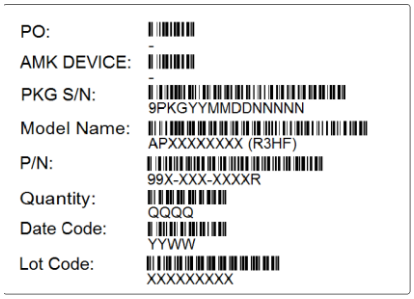
Label A→ Anti-static and humidity notice



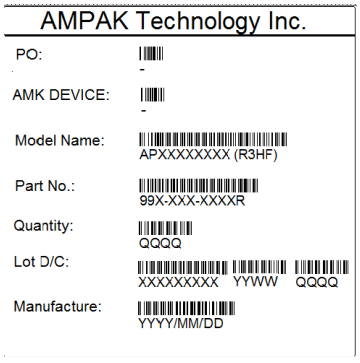

Label B→ MSL caution / Storage Condition



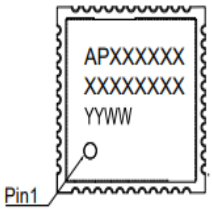

Label C→ Inner box label .




Label D→ Carton box label .

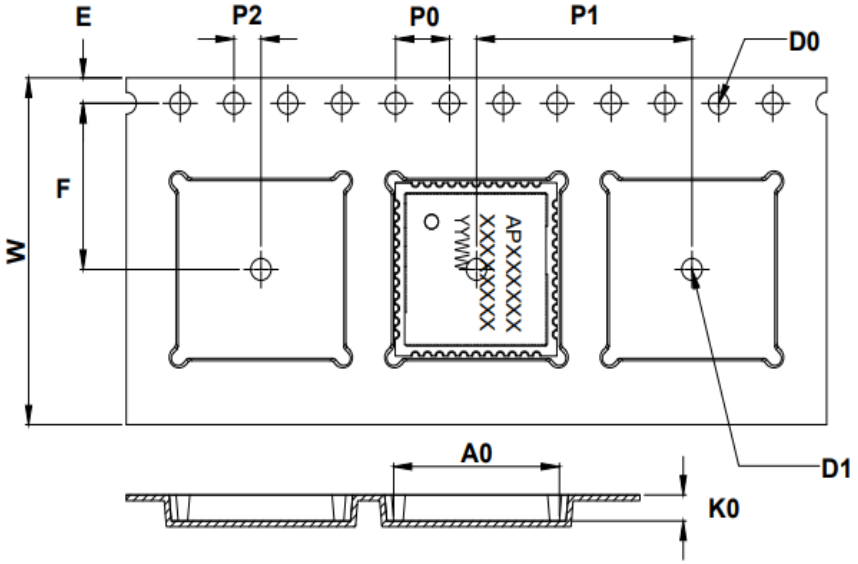


# 13.2 Dimension



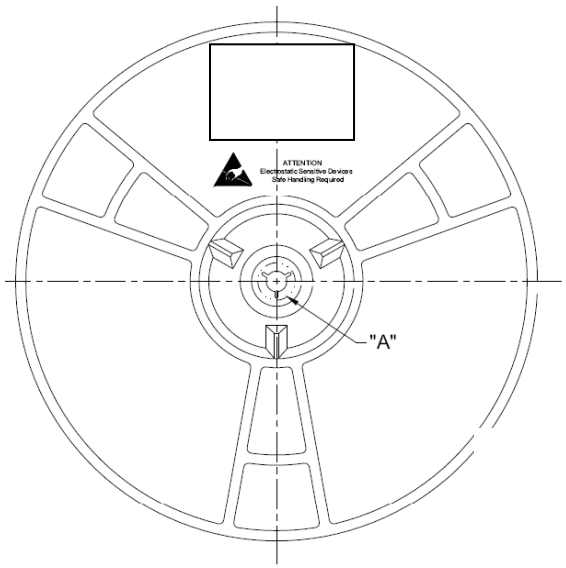

| Rows | Content | Note |
|---|---|---|
| Row 1 | APXXXXXX | Model Name |
| Row 2 | XXXXXXXX or XXXXXXXXX<br>( 8 digitals or 9 digitals ) | Lot Code |
| Row 3 | YYWW | Date Code |
| Row 4 | Through hole or non through hole | Pin1 |

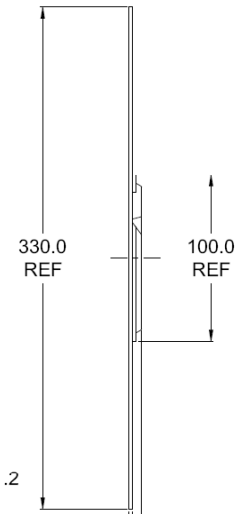
| Packaging Type | Packing Quantity | Note |
|---|---|---|
| Reel / Bag / Box | 1500pcs | 13" reel |
| Carton | 7500pcs | 5Box |

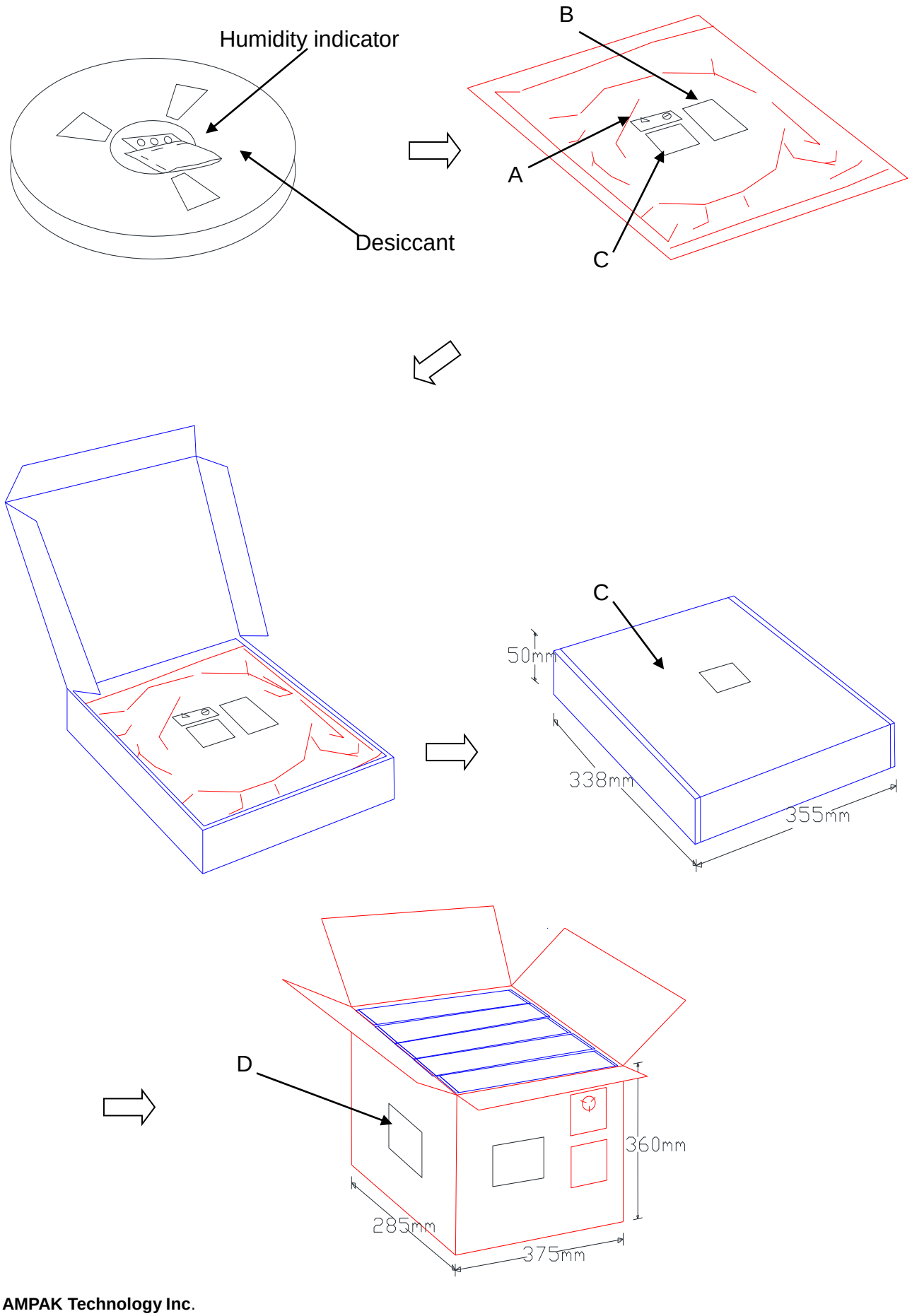



| | |
|---|---|
| W | 24.00±0.30 |
| A0 | 12.30±0.10 |
| B0 | 12.30±0.10 |
| K0 | 1.80±0.10 |
| E | 1.75±0.10 |
| F | 11.50±0.10 |
| P0 | 4.00±0.10 |
| P1 | 16.00±0.10 |
| P2 | 2.00±0.10 |
| D0 | 1.50 +0.10 -0.00 |
| D1 | ϕ1.50MIN |

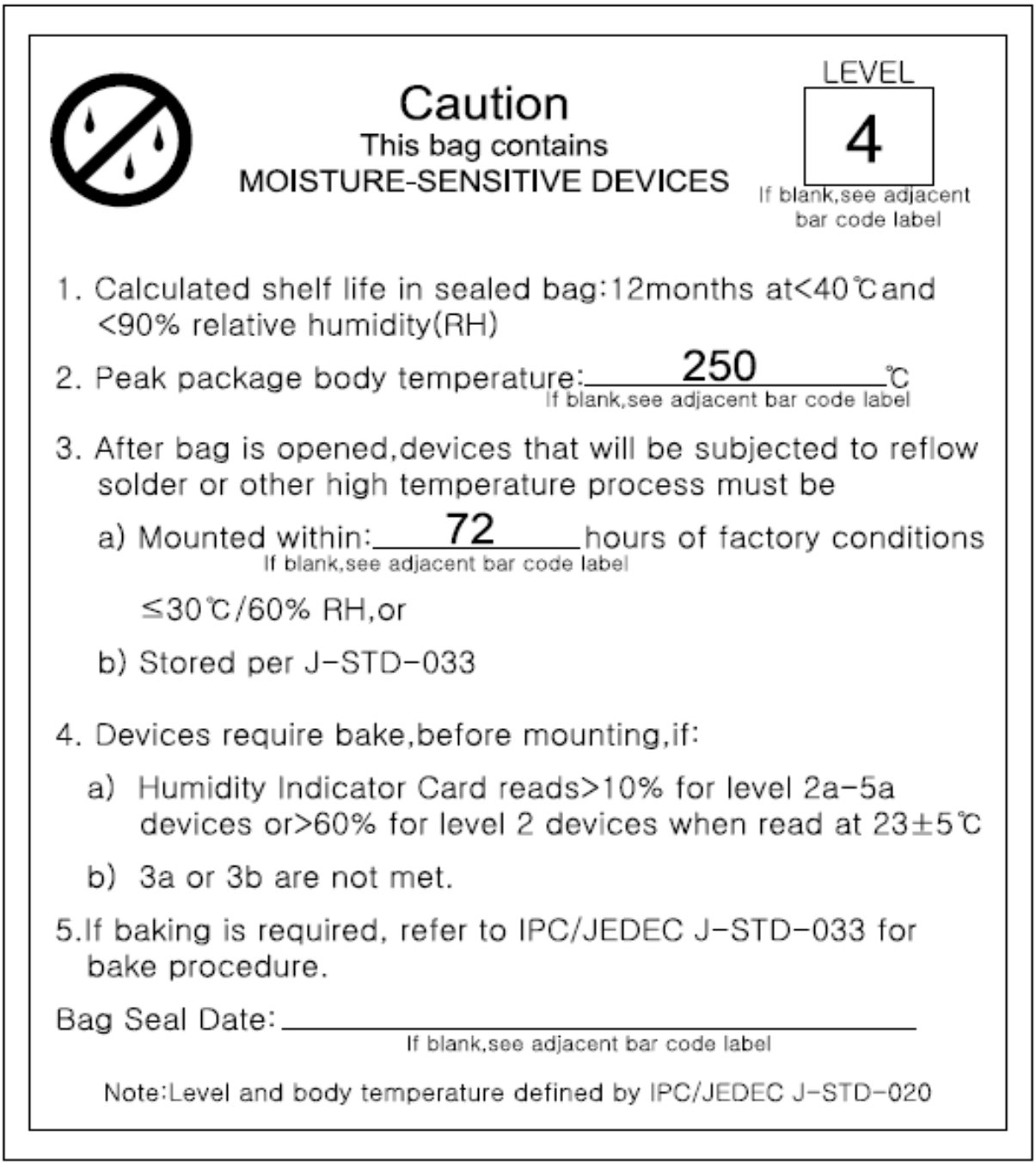
1. Material : Black Conductive Polystyrene Alloy.
2. All dimensions meet EIA-481-D requirements.
3. Thickness : 0.30±0.05mm.




Humidity indicator
Desiccant
B
A
C
C
50mm
338mm
355mm
D
360mm
285mm
375mm

## 13.3 MSL Level / Storage Condition



**Caution**
This bag contains
MOISTURE-SENSITIVE DEVICES

LEVEL
4
If blank,see adjacent bar code label

1. Calculated shelf life in sealed bag:12months at<40℃and <90% relative humidity(RH)
2. Peak package body temperature:250℃
   If blank,see adjacent bar code label
3. After bag is opened,devices that will be subjected to reflow solder or other high temperature process must be
   a) Mounted within:72 hours of factory conditions
      If blank,see adjacent bar code label
      ≤30℃/60% RH,or
   b) Stored per J-STD-033
4. Devices require bake,before mounting,if:
   a) Humidity Indicator Card reads>10% for level 2a-5a devices or>60% for level 2 devices when read at 23±5℃
   b) 3a or 3b are not met.
5. If baking is required, refer to IPC/JEDEC J-STD-033 for bake procedure.

Bag Seal Date: _____
If blank,see adjacent bar code label

Note:Level and body temperature defined by IPC/JEDEC J-STD-020